

Homework 5

In this homework, you will revisit questions on Exam 1 and make use of Matlab and Simulink to analyze the systems in more detail.

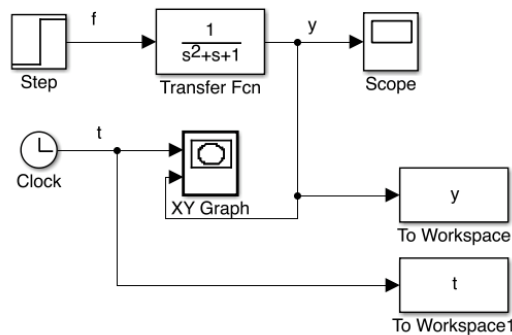
Question 1

- Recreate the system in Matlab or Simulink.
- For various positive values of k , show the effect of a unit step disturbance by plotting the system output vs. time (ideally all on the same graph with a legend).
- Do your analytical solutions on the Exam (for steady-state value, time constant) match what you see in the simulations?

Question 2

- Recreate the model shown in Question 2 in Simulink.
- On the plot displayed on the scope, label the system specifications listed in the bullet points.

Note: If you want “nicer” plots, you can use the XY Graph feature, or send both t and y out to the workspace (under “Sinks”) and use the Matlab `plot(t,y)` command. For a mass-spring-damper system with no feedback, this would look something like this:



- Calculate the final value of y analytically if the input to the system is a step of magnitude k_1 ($f(t) = k_1\mu(t)$). Show that your Simulink model gives the same results for a few values of k_1 .
- Show that the rise time, settling time, % overshoot, and peak time are not affected by k_1 using your Simulink model. Prove this analytically as well.

Question 3

- Define a transfer function $G(s)$ that has 2 imaginary poles $\{+2j, -2j\}$ using `G = zpkl(...)`. See our code examples if you forgot how to do this. Show the impulse response over a period of 10 seconds using `impz(G, 10)`. Explain the response, and compare to the analytical solution (calculate the period and frequency analytically and compare).

- f. Define a transfer function $G(s)$ that has 2 real poles $\{-1, -2\}$. Determine the analytical response to an impulse. Compare your result to `impz(G, 10)`.
- g. Define a transfer function $G(s)$ that has 2 complex conjugate poles $(-1 \pm j)$ in the left half plane. Move the poles to the left ($G_2(s)$) and further away from the x-axis ($G_3(s)$). Plot the impulse response for all 3 cases on the same figure and explain the output.

Question 5

- h. Recreate the system in Simulink.
- i. Use the value of K calculated in the exam that makes the rise time less than 1 and show the output on the scope. Is it what you expected?
- j. What happens as K increases? Show an example output and qualitatively explain the changes (explain changes in rise time and overshoot).
- k. For part (b), there were several ways to estimate the value of K that results in the plot displayed. Using the value of K you calculated on your exam, show the scope output. How close were you?

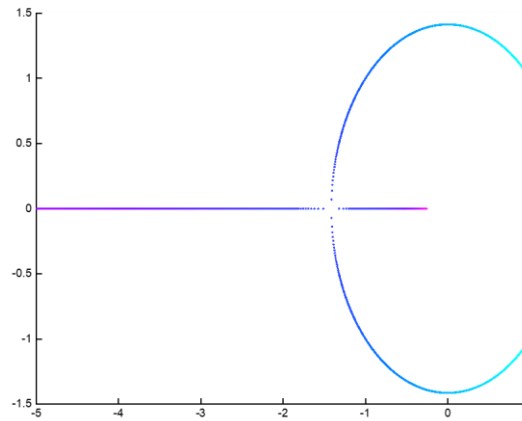
Question 7

- l. Using Matlab (not Simulink), define G_1 and G_2 . Use the commands `impz()`, `step()`, and `lsim()` to recreate the 6 plots. Review our code examples if you forget how to apply an arbitrary input using `lsim()`.

Question 8

- m. Use the functions `pzmap()` and `pzplot()` to calculate the poles and zeros of G and plot them in the complex plane.
- n. Vary K from 0 to 10 (in increments of .2 or smaller). Calculate the 2 poles of G_2 for each value of K (Use a loop to make this easy).
- o. Plot the poles as a function of K on the complex plane. Label 3 regions – (1) when the poles are complex and on the right-half plane, (2) when the poles are complex and on the left-half plane, and (3) when the poles are real (on the x-axis). On your plot, include the value of K and values for the poles that divide the regions. Draw arrows showing the direction of how the poles move for increasing K .

Hint: your plot should look something like this:



- p. Calculate the boundaries of the 3 regions analytically by considering that a polynomial of the form $ax^2 + bx + c = 0$ has roots

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

and examining when $b^2 - 4ac > 0$, for example.

- q. Show that as K goes to infinity, the real poles stay on the left-half plane by (i) using a very large value of K in your code and (ii) proving analytically. What are the 2 poles at this limit?
- r. Consider the region where the poles are complex conjugates on the LHP. As K increases, what happens to the damping ratio, natural frequency, and damped natural frequency? What is happening to the % overshoot, rise time, and settling time?
- s. Consider the region where the poles are real on the LHP. What happens as K increases? Remember, there are 2 first-order poles, so there will be two time constants. Does increasing K decrease the overall settling time of the system?
- t. Consider all regions. What value of K gives the shortest settling time overall? Show this by plotting the system response to a unit step for this value of K and for values of K above and below this value on the same plot.